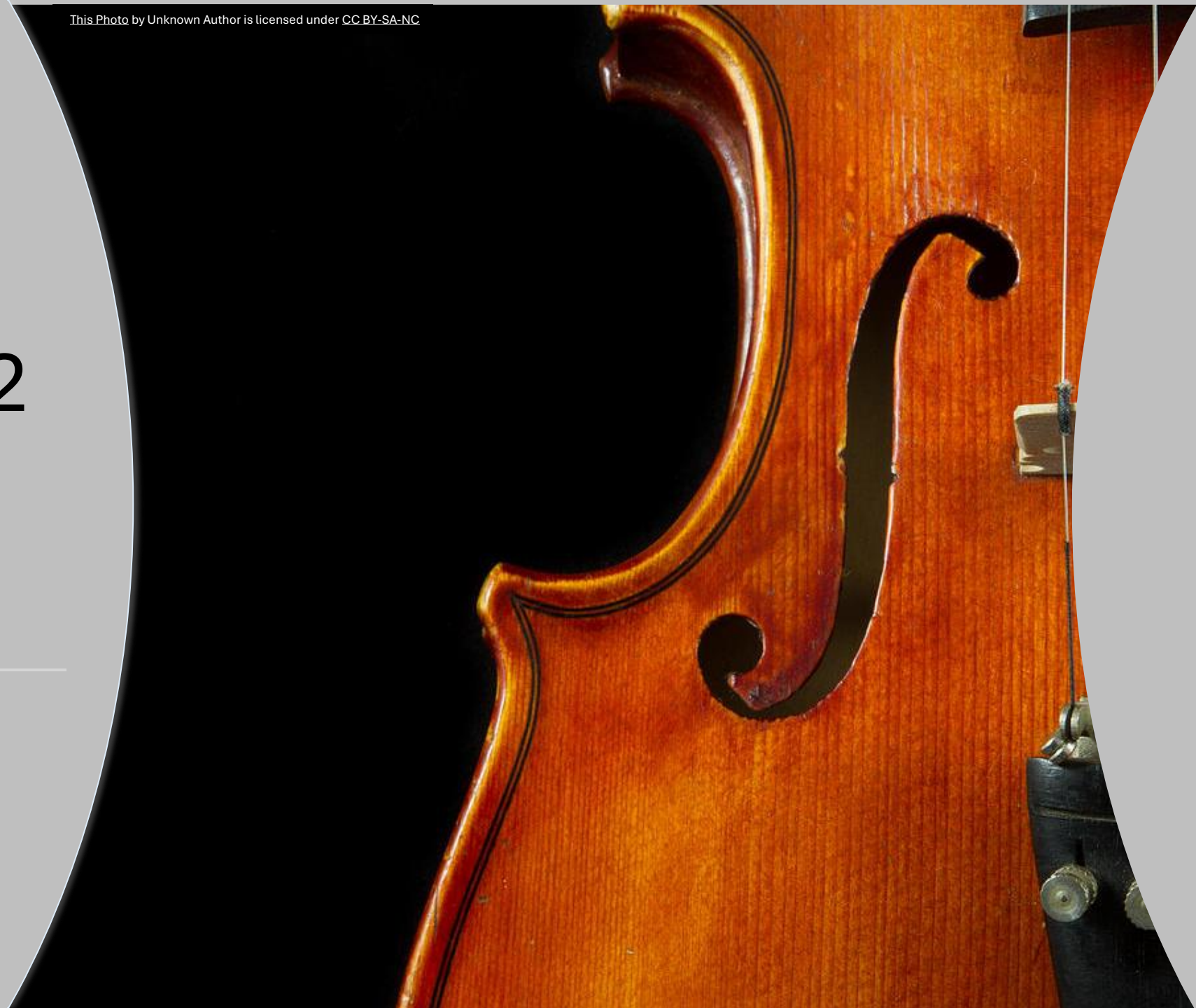


Mathematics 2 (5EZB0) Lecture 9

Koondi Mitra

*Mathematics & Computer Science
Department*



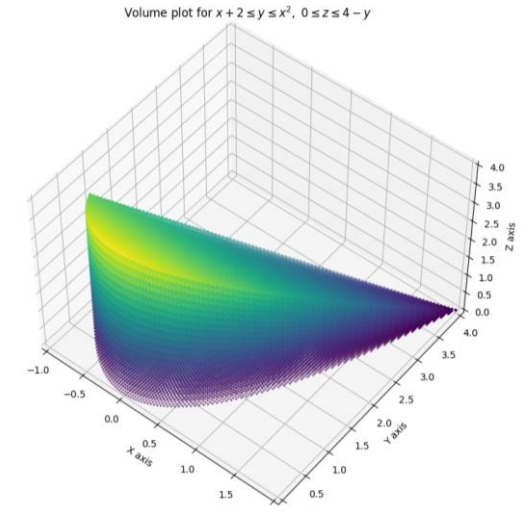
Triple integrals

$$\iiint_{\mathcal{D}} f(x, y, z) dx dy dz$$

Run **volume_integral.py** to find the volume and mass of a sphere using voxels

- Volume of domain $\mathcal{D}$: $\int \int \int_{\mathcal{D}} dx dy dz$, Mass of solid: $\int \int \int_{\mathcal{D}} \rho(x, y, z) dx dy dz$
- Satisfies *additivity, linearity, positivity, order preservation, and triangle inequality*
- For $\mathcal{D} := \{(x, y, z): a \leq x \leq b, c(x) \leq y \leq d(x), e(x, y) \leq z \leq g(x, y)\}$

$$\int \int \int_{\mathcal{D}} f(x, y, z) dx dy dz = \int_a^b \left(\int_{c(x)}^{d(x)} \left(\int_{e(x, y)}^{g(x, y)} f(x, y, z) dz \right) dy \right) dx$$



- If f is continuous, and an integral is well defined in two separate orders, then the order of integration can be exchanged
- Higher order integrations are also written simply as $\int_{\mathcal{D}} f dx$

Questions 1

Find the volume of the region
 $\mathcal{V} := \{x + 2 \leq y \leq x^2, 0 \leq z \leq 4 - y\}$

Change of variables

Now consider the change of variable

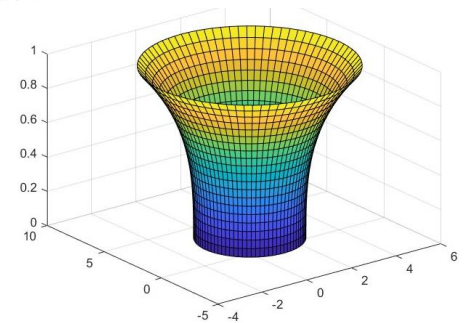
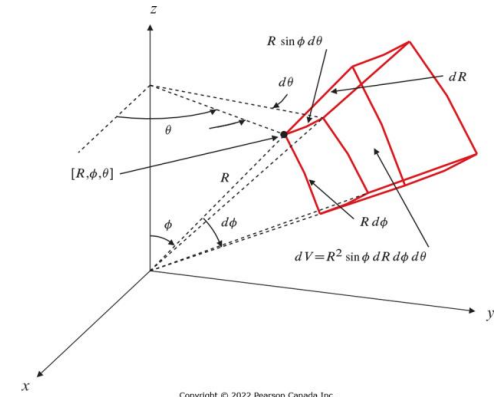
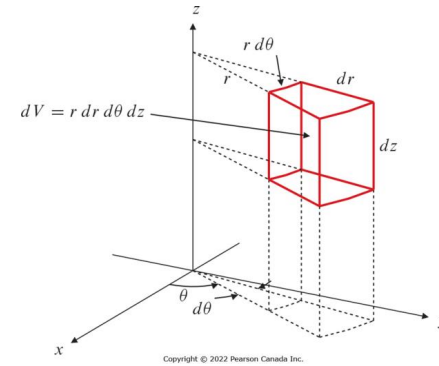
$$\begin{aligned}x &= x(u, v, w) \\ y &= y(u, v, w) \\ z &= z(u, v, w)\end{aligned}$$

➤ Let $f(x(u, v, w), y(u, v, w), z(u, v, w)) = h(u, v, w)$

➤ Let the domain $\mathcal{D}$ be transformed into S in (u, v, w) coordinates

Then, the area element changes by $dx dy dz = \left| \det \left(\frac{\partial(x, y, z)}{\partial(u, v, w)} \right) \right| du dv dw$

$$\iiint_{\mathcal{D}} f(x, y, z) dx dy dz = \iiint_S h(u, v, w) \left| \det \left(\frac{\partial(x, y, z)}{\partial(u, v, w)} \right) \right| du dv dw$$



Questions 2.1

Find the volume of the curve between $z = 0$ and 1

$$\frac{(x-1)^2}{a^2} + \frac{(y-2)^2}{b^2} = z^3 + 1$$

Change of variables: cylindrical coordinate

Now consider the change of variable

$$x = r \cos \theta$$

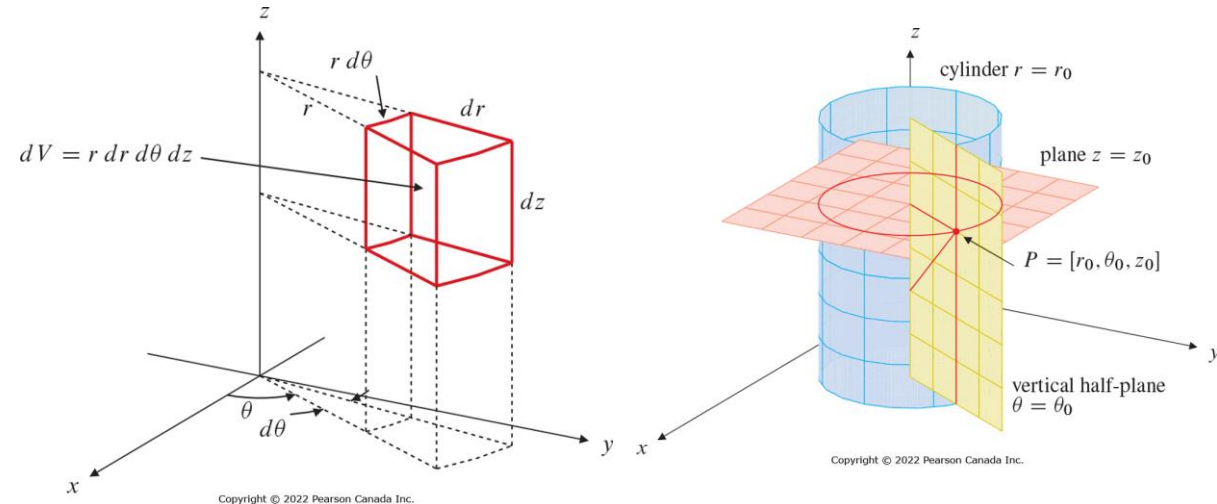
$$y = r \sin \theta$$

$$z = z$$

which gives

$$\frac{\partial(x, y, z)}{\partial(r, \theta, z)} = \begin{pmatrix} \cos \theta & -r \sin \theta & 0 \\ \sin \theta & r \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\text{And } \left| \det \left(\frac{\partial(x, y, z)}{\partial(r, \theta, z)} \right) \right| = r$$



Questions 3

Find the coordinates of the center of mass

$$\bar{x}_i = (\int_{\mathcal{D}} x_i \rho dx) / (\int_{\mathcal{D}} \rho dx) \text{ of the cone}$$

$$\mathcal{D} := \{z = \sqrt{x^2 + y^2}, 0 \leq z \leq h\}$$

Change of variables: spherical coordinate

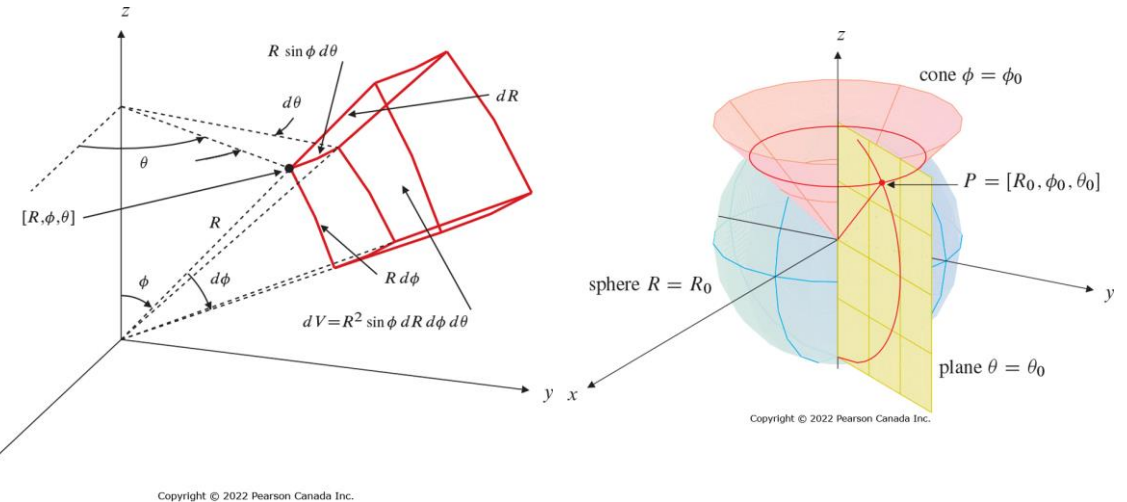
Now consider the change of variable

$$\begin{aligned}x &= R \sin \phi \cos \theta \\y &= R \sin \phi \sin \theta \\z &= R \cos \phi\end{aligned}$$

which gives

$$\frac{\partial(x, y, z)}{\partial(R, \phi, \theta)} = \begin{pmatrix} \sin \phi \cos \theta & R \cos \phi \cos \theta & -R \sin \phi \sin \theta \\ \sin \phi \sin \theta & R \cos \phi \sin \theta & R \sin \phi \cos \theta \\ \cos \phi & -R \sin \phi & 0 \end{pmatrix}$$

And $\left| \det \left(\frac{\partial(x, y, z)}{\partial(R, \phi, \theta)} \right) \right| = R^2 \sin \phi$



Questions 4.1

Find the volume inside the sphere $x^2 + y^2 + z^2 = a^2$ and the cone $z = \sqrt{x^2 + y^2}$? Find also the formula of the volume of a sphere.

Questions 4.2

Find the integral

$$\int_{\mathbb{R}^3} e^{-x^2-2y^2-3z^2} dx dy dz$$

Surface area

We saw earlier that a surface normal of $z = f(x, y)$ is at

$$\mathbf{n} = (\nabla f(x, y), -1)$$

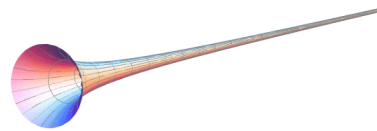
➤ The angle with the z -axis is then $\cos \gamma = \frac{|\mathbf{n} \cdot \mathbf{k}|}{|\mathbf{n}| |\mathbf{k}|} = \frac{1}{\sqrt{1 + |\nabla f|^2}}$

➤ Then the area of the surface element would be

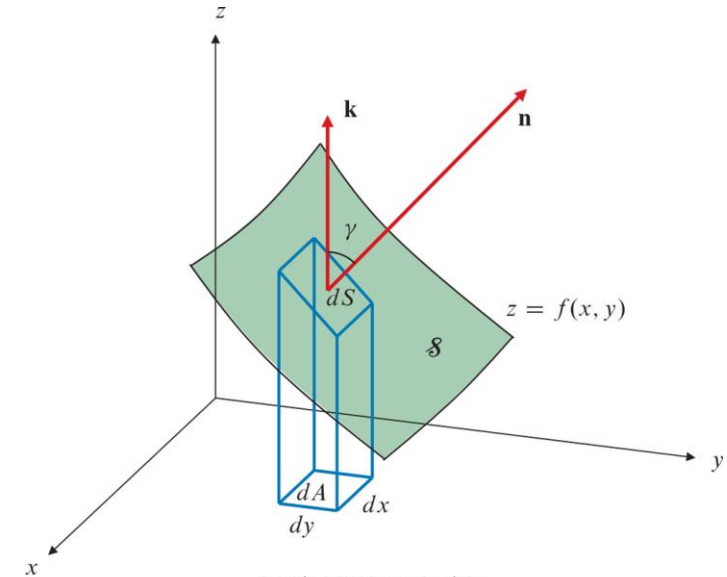
$$dS = \frac{dA}{\cos \gamma} = \sqrt{1 + |\nabla f|^2} dA$$

➤ Then the surface area is

$$S = \iint_{\mathcal{D}} \sqrt{1 + |\nabla f|^2} dA$$



This Photo by Unknown Author is licensed under [CC BY-SA-NC](#)



Copyright © 2022 Pearson Canada Inc.

Run **surface_integral.py** to find the tessellation of surfaces

Questions 5.1

Find the area of the surface of a sphere

Questions 5.2 (surface of revolution)

Show that the surface area of the function $z = f(r)$ ($r > 0$) is $2\pi \int_0^\infty r \sqrt{1 + f'(r)^2} dr$

Questions 5.3 (Gabriel's horn)

What is the surface area and the volume over $\{z > 0\}$ of the function $z = \frac{1}{r}$ in $B_1((0,0)) = \{r < 1\}$?

Revisiting Scalar & Vector fields

Vector fields

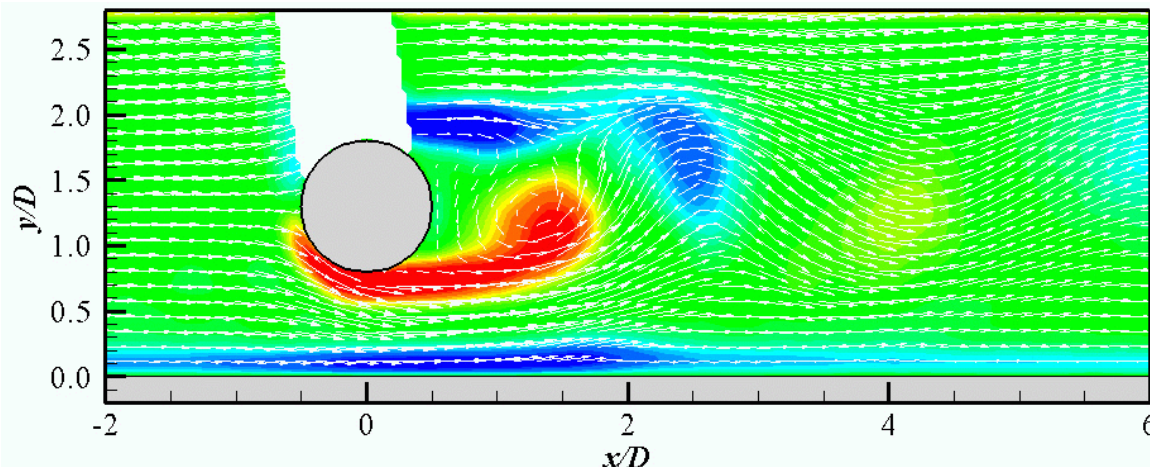
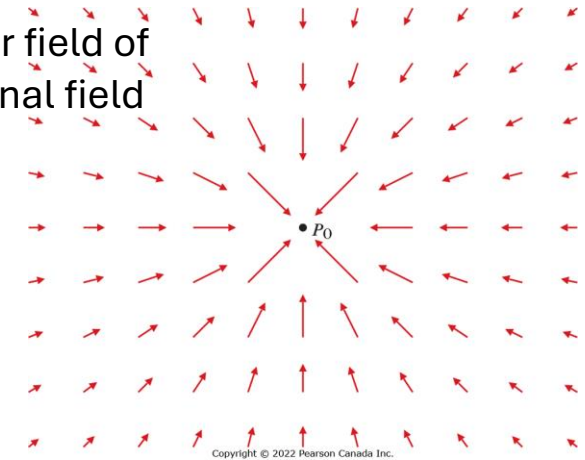
For a domain $\mathcal{D} \subseteq \mathbb{R}^d$, a vector field $\mathbf{f}$ is a function of the form

$$\mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_n(\mathbf{x}))$$

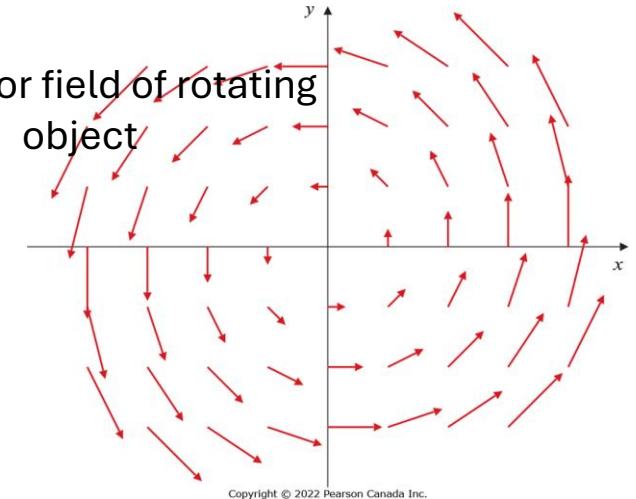
written also as

$$\mathbf{f}: \mathcal{D} \rightarrow \mathbb{R}^n$$

The vector field of
gravitational field



The vector field of rotating
object



Field lines

Field lines

For the vector field $\mathbf{f}: \mathcal{D} \rightarrow \mathbb{R}^d$ its field-lines/ stream-lines are family of curves $\mathbf{r}(t) = \sum_{i=1}^d x_i \hat{\mathbf{e}}_{x_i}$ described by parameter t , such that on any point of the curve, the field $\mathbf{f}$ is tangent to the curve. i.e.,

$$\frac{d\mathbf{r}}{dt}(t) = \lambda(t)\mathbf{f}(\mathbf{r}(t))$$

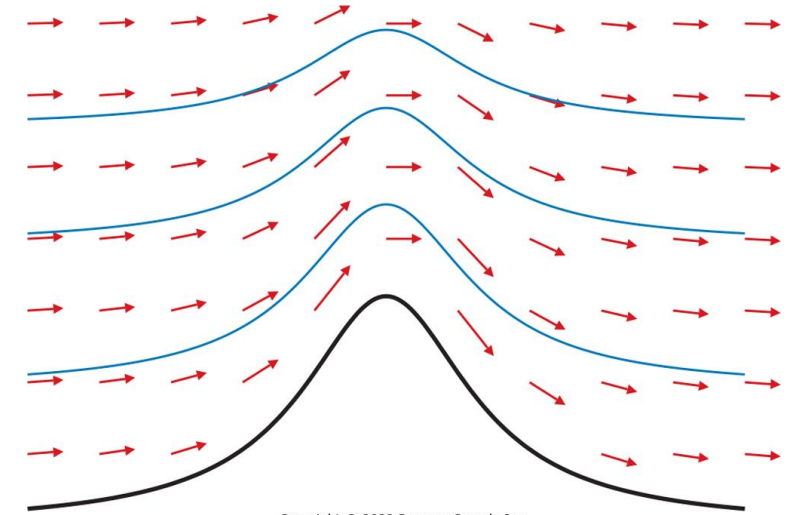
This yields the equation for all $1 \leq i \leq d$

$$\frac{dx_i}{f_i(\mathbf{r})} = \lambda(t)dt$$

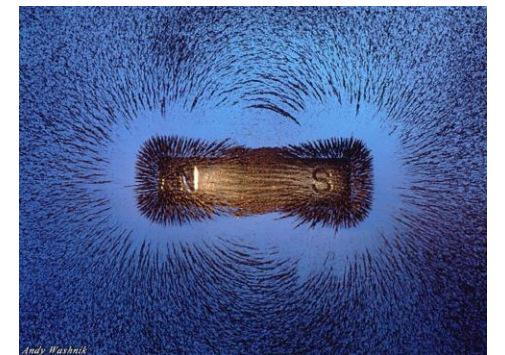
Question 6

What are the field lines of

1. The electric field of a point-charge?
2. A rotating object?



Copyright © 2022 Pearson Canada Inc.



This Photo by Unknown Author is licensed under [CC BY-SA](#)